

Katie Li

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Education

University of Illinois at Urbana-Champaign

Expected Graduation: May 2027

Bachelor of Science in Computer Science (GPA: 4.0/4.0)

Champaign, IL

- **Honors:** 2025 Illinois Engineering Achievement Award | Rothmuller Award | Dean's List
- **Coursework:** Algorithms & Models of Computation, Data Structures, Computer Architecture, Software Design

Technical Skills

Languages: C++, Java, Python, JavaScript/TypeScript, SQL

Frameworks & Tools: AWS, Docker, Git, Flask, React, PostgreSQL, HTML/CSS, TensorFlow, PyTorch

Certifications: Machine Learning Specialization (DeepLearning.AI, Stanford)

Experience

PrairieLearn

Sep 2025 – Present

Software Developer

Champaign, IL

- Built containerized production components in **Docker** and interactive modules using **Python**, **HTML**, and **JavaScript** for the University of Illinois' online learning/testing platform.
- Deployed features across core CS courses (CS 374, 225), used by over **14,000 students** annually.
- Wrote **pytest** unit tests to validate question-generation and grading logic, ensuring reliable course assessments.

University of Illinois (Forward Data Lab)

Aug 2025 – Present

Machine Learning Researcher

Champaign, IL

- Built a scalable **retrieval-augmented generation (RAG)** system with **vector search** and **LLM integration**, enabling efficient literature discovery in **NLP research**.
- Engineered **Python** retrieval pipelines over **320k+ arXiv papers** by querying data from the arXiv **REST API** and generating embeddings with **FAISS** and transformers.

Penn State University (College of IST)

May 2025 – Jul 2025

Research Assistant

University Park, PA

- Designed a multi-modal AI pipeline to evaluate instructional videos, under submission to **WACV 2026**.
- Wrote **Python batch processing pipelines** integrating **LLM prompts** and **OpenAI API** calls, automating and producing model evaluations and utilizing **Whisper** for audio extraction and **GPT-4o** for analysis.
- Scaled the system to process **400 hours** of content with **80% evaluator alignment** in experimental trials.

Robotics For All

Jan 2022 – Jul 2022

Web Development Intern

Cupertino, CA

- Revamped production website features in **React** and **JavaScript**, supporting nonprofit outreach to **3,000 users**.
- Deployed a new donations workflow, integrating **Google Forms** with **backend services** to streamline fundraising.

Projects

Bookrates: Book Recommendation Platform | React, Flask, AWS, Python, JavaScript, CSS

- Built and deployed a **full-stack** website with a **React** frontend and **Flask** backend hosted on **AWS EC2**.
- Developed **REST API** endpoints to **process recommendation queries** and return ranked JSON results, integrating with a **database of 70k+ books** and **1M+ user interactions**.
- Optimized system performance through adding Parquet-based caching layers, reducing query times by **2x**.

Stride: Mobile Fitness App | TypeScript, React Native, PostgreSQL, XCode, TensorFlow

- Built a cross-platform fitness app using **React Native** + **TypeScript**, with run tracking features and progress graphs.
- Implemented a **PostgreSQL backend** with **Supabase**, supporting user authentication and credential storage.
- Integrated **MoveNet** for real-time inference, enabling instant **AI powered pose detection** and analysis.

Leadership

MVHS Competitive Programming Club

Aug 2021 – Jun 2024

Director · USACO Gold Tier Competitor

Cupertino, CA

- Achieved **USACO Gold Division** status by solving advanced competitive programming problems in **C++**.
- Directed a **30-member club** and led **80+ technical workshops**, growing membership by **50%** over two years.